Last Meeting (6 February)

What we discussed:

- Talked about cross-selling
- Presented a model of switching costs with persistence in preferences

Today's meeting:

- Economics of policies that change lenders' information sets
- Data we would have access to
- Structural model for estimation

Policies that affect the amount of public information for lenders

Internationally:

- Fair Credit Reporting Act (FCRA) sets Time limits (e.g. 10 years for bankruptcies)
- Open Banking Policies (e.g. in the UK) require banks to share customer data (including balances and transactions) with third parties

In Chile:

- Restrictions to credit bureaus information. They deleted delinquencies for short term unemployed individuals, excluded credit inquiries from credit scores and once they erased default histories for amounts below $1,500.
- Expand the reporting entities: incorporated credit cards by retailers (law 21,130) and incorporated other non-bank lenders (law 21,680)

Economics of changing lenders' information sets

- **Setup:** Incumbent bank has private info on borrowers, rival banks do not
- Effects of policies that expand public information:
  1. **Reduces adverse selection:** switchers are no longer assumed to be lemons
  2. **Reduces incumbent's information rents:** Incumbent's advantage came from knowing h (default probability) when rivals did not. If rivals can also screen, the "winner's curse" weakens $\Rightarrow$ incumbent's market power falls
  3. **Affects negatively risky borrowers:** uninformed lenders may raise their rates or deny them credit

Predictions:

P1. Switching rates increase for good borrowers (rival banks price more aggressively)

P2. Interest rates fall for good borrowers (separating $\rightarrow$ competitive pricing)

P3. Default rates of switchers decrease (reduction of adverse selection)

P4. Negative effect on risky borrowers: may face higher rates or lose access

Data: Chilean Administrative Credit Records (CMF)

Individual-level data on:

- **Credit cards** (bank & non-bank): usage, payments, delinquency
- **Loans:** amounts, interest rates, payment plans, maturities, delinquency
- **Accounts** (checking, savings): balances and transactions

Unit of observation: borrower $\times$ lender $\times$ month.

Structural Model: Two-Period Game

Period 1 — First loan:

- Two banks compete on rate r_{j1}; neither observes borrower type $h \sim F(h)$
- Winning bank learns h from the relationship; borrower develops switching cost λ

Period 2 — Second loan:

- Incumbent sets rate $r_{12} = \sigma^*(h)$ and rival sets $r_{22} = r_2^*$
- Borrower chooses via demand: $s(r_{12}, r_{22}) = \dfrac{1}{1 + \exp(\alpha(r_{12} - r_{22}) - \lambda)}$

Backward induction: Period-2 equilibrium yields continuation values $V^I > V^R > 0$. In Period 1, each bank solves:

$$\max_{r_{j1}} \; s(r_{j1}, r_{-j1}) \cdot \Big[E_h[(1-h)\, r_{j1} - 1] + \Delta V \Big], \qquad \Delta V = V^I - V^R$$

Banks price Period-1 loans below break-even to secure the incumbent position in Period 2.

Structural Model: Period-2 Equilibrium

Incumbent (bank 1) prices each type at break-even + markup:

$$\sigma^*(h) = \underbrace{\frac{1}{1-h}}_{\text{break-even}} + \underbrace{\frac{1}{\alpha\big(1 - s(\sigma^*(h), r_2^*)\big)}}_{\text{markup}}$$

Rival (bank 2) sets a single rate:

$$r_2^* = \frac{1}{1-\tilde{h}} + \frac{1}{\alpha} \cdot \frac{\int (1-s)(1-h)\, dF}{\int s\,(1-s)(1-h)\, dF}$$

Primitives to estimate: α (price sensitivity), λ (switching cost), $F(h)$ (type distribution)

Structural Model: Identification

- Observables: winning rate r_i^{win}, winning bank $W_i \in \{1, 2\}$, default $D_i \in \{0, 1\}$
- Sequential identification:
  1. **Rival rate:** r_2^* identified as the common rate among bank-2 winners
  2. **Type recovery:** $E[D_i \mid r_i^{\text{win}} = r,\ W_i = 1] = h \quad \Rightarrow$ invert σ^* to recover each borrower's type
  3. α **and** λ**:** From the equilibrium FOC, for any two types $h_a \neq h_b$:

     $$\ln\big(\alpha\, m(h) - 1\big) \;=\; \lambda \;-\; \alpha\, \Delta(h)$$

     where $m(h) = \sigma^*(h) - \frac{1}{1-h}$ is the markup and $\Delta(h) = \sigma^*(h) - r_2^*$ is the rate gap. Slope identifies α; intercept identifies λ.
  4. **Type distribution:** $F(h)$ recovered from bank-1 winners, correcting for selection via $s(h)$